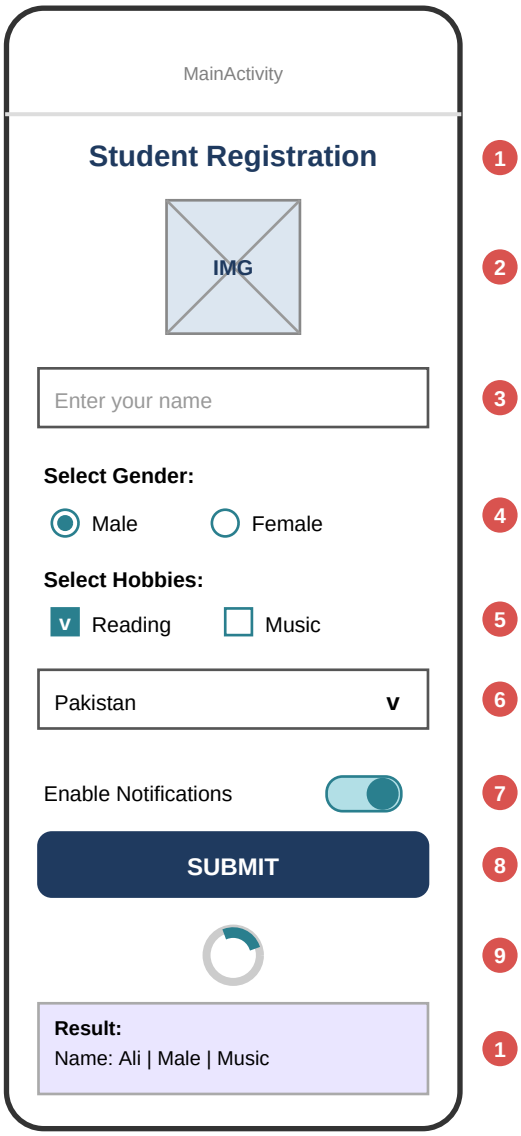
Ek Screen par 9 Views

TextView, ImageView, EditText, RadioButton, CheckBox, Spinner, Switch, Button, ProgressBar | XML + Kotlin + code likhne ka tareeqa

Neeche ek aisi screen hai jis mein aap ke bataye hue **saare 9 views** ek hi jagah hain. Pehle **tasveer** dekho (har view par laal number hai), phir har view ka matlab, phir poora **XML aur Kotlin code**, aur aakhir mein **code likhne ka tareeqa**.

1. Screen ki tasveer



1	**TextView**	Title aur result likhne wali jagah
2	**ImageView**	Tasveer / logo
3	**EditText**	Naam likhne ki jagah
4	**RadioButton**	Male / Female (sirf ek)
5	**CheckBox**	Hobbies (kayi chun sakte hain)
6	**Spinner**	Drop-down se mulk chunna
7	**Switch**	Notifications on / off
8	**Button**	Submit dabane ki cheez
9	**ProgressBar**	Ghoomta hua 'loading' gola

Screen par user ka kaam: naam likhe, gender aur hobbies chune, mulk chune, notifications on/off kare, **Submit** dabaye. ProgressBar thori der ghoomta hai, phir neeche **Result** mein sab kuch likha aa jata hai.

2. Har view kya hai? (aasan zubaan mein)

View	Aasan matlab	XML tag	Kotlin mein use
TextView	Sirf **likha hua** dikhata hai. User badal nahi sakta. Jaise board par likha naam.	<TextView>	tv.text = "hello"

View	Aasan matlab	XML tag	Kotlin mein use
ImageView	**Tasveer** dikhata hai (logo, photo). Tasveer res/drawable mein rakhte hain.	<ImageView>	android:src="@drawable/..."
EditText	**Khali dibba** jis mein user likhta hai. hint = halki hidayat.	<EditText>	et.text.toString()
RadioButton	Gol button; group mein se **sirf ek** chun sakte hain. RadioGroup mein rakhna lazmi.	<RadioGroup> + <RadioButton>	rb.isChecked
CheckBox	Chaukor tick dibba; **kayi** ek saath chun sakte hain.	<CheckBox>	cb.isChecked
Spinner	**Drop-down list**; click par options khulte hain, ek chunte hain.	<Spinner>	sp.selectedItem.toString()
Switch	Bijli ka **on/off** switch.	<Switch>	sw.isChecked
Button	**Dabane** wali cheez; dabane par kaam shuru.	<Button>	btn.setOnClickListener { }
ProgressBar	'**Kaam ho raha hai**' dikhane wala ghoomta gola.	<ProgressBar>	pb.visibility = View.VISIBLE / GONE

Yaad rakho: CheckBox = kayi chun sakte hain. RadioButton = sirf ek (RadioGroup ke saath). Switch = on/off. Spinner = list se ek.

3. Poora XML code (activity_main.xml)

Screen mein 9 views hain jo ek chhoti screen mein na aayen to neeche ka hissa kat jata hai. Isliye sab se bahar **ScrollView** lagaya hai (scroll ho kar sab dikhe). ScrollView mein sirf **ek** hi child ho sakta hai, isliye uske andar ek **LinearLayout** hai jis mein saare views hain.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- 1. TextView (title) -->
    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Student Registration"
        android:textSize="22sp"
        android:textStyle="bold"
        android:gravity="center" />

    <!-- 2. ImageView -->
    <ImageView
        android:id="@+id/imgLogo"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_gravity="center"
        android:src="@drawable/ic_launcher_foreground"
        android:contentDescription="Application logo" />

    <!-- 3. EditText -->
    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:inputType="textPersonName" />

    <!-- 4. RadioGroup + RadioButtons -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Select Gender:"
        android:textStyle="bold" />

    <RadioGroup
        android:id="@+id/rgGender"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <RadioButton
            android:id="@+id/rbMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male"
            android:checked="true" />

        <RadioButton
            android:id="@+id/rbFemale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female" />
    </RadioGroup>
```

```
    <!-- 5. CheckBoxes -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Select Hobbies:"
        android:textStyle="bold" />

    <CheckBox
        android:id="@+id/cbReading"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Reading" />

    <CheckBox
        android:id="@+id/cbMusic"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Music" />

    <!-- 6. Spinner -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Select Country:"
        android:textStyle="bold" />

    <Spinner
        android:id="@+id/spCountry"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <!-- 7. Switch -->
    <Switch
        android:id="@+id/switchNotify"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Enable Notifications"
        android:checked="true" />

    <!-- 8. Button -->
    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="SUBMIT" />

    <!-- 9. ProgressBar (shuru mein chhupa hua) -->
    <ProgressBar
        android:id="@+id/progressBar"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:visibility="gone" />

    <!-- Result TextView -->
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:padding="12dp"
```

```
        android:text="Result:"
        android:textSize="15sp" />

</LinearLayout>
</ScrollView>
```

Dhyan do: har view ka **android:id** alag hai (tvTitle, imgLogo, etName, rgGender, rbMale, rbFemale, cbReading, cbMusic, spCountry, switchNotify, btnSubmit, progressBar, tvResult). Kotlin mein inhi naamon se views ko pakra jata hai. ProgressBar par **visibility=gone** isliye ke shuru mein chhupa rahe, Submit dabane par dikhe.

4. Poora Kotlin code (MainActivity.kt)

MainActivity.kt:

```
package com.example.allviewsapp

import android.os.Bundle
import android.os.Handler
import android.os.Looper
import android.view.View
import android.widget.ArrayAdapter
import android.widget.Button
import android.widget.CheckBox
import android.widget.EditText
import android.widget.ImageView
import android.widget.RadioButton
import android.widget.Spinner
import android.widget.Switch
import android.widget.TextView
import android.widget.ProgressBar
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)          // XML jodo

        // ---- Step 1: saare views pakro (findViewById) ----
        val etName = findViewById<EditText>(R.id.etName)
        val rbMale = findViewById<RadioButton>(R.id.rbMale)
        val rbFemale = findViewById<RadioButton>(R.id.rbFemale)
        val cbReading = findViewById<CheckBox>(R.id.cbReading)
        val cbMusic = findViewById<CheckBox>(R.id.cbMusic)
        val spCountry = findViewById<Spinner>(R.id.spCountry)
        val switchNotify = findViewById<Switch>(R.id.switchNotify)
        val btnSubmit = findViewById<Button>(R.id.btnSubmit)
        val progressBar = findViewById<ProgressBar>(R.id.progressBar)
        val tvResult = findViewById<TextView>(R.id.tvResult)
        val imgLogo = findViewById<ImageView>(R.id.imgLogo)

        // ---- Step 2: Spinner mein list bharo (adapter) ----
        val countries = arrayOf("Pakistan", "Qatar", "UAE", "Saudi Arabia")
        val adapter = ArrayAdapter(this,
            android.R.layout.simple_spinner_item, countries)
        adapter.setDropDownViewResource(
            android.R.layout.simple_spinner_dropdown_item)
        spCountry.adapter = adapter

        // ---- Step 3: Button click par kaam ----
        btnSubmit.setOnClickListener {
            val name = etName.text.toString()              // EditText
            if (name.isEmpty()) {
                tvResult.text = "Please enter your name"
                return@setOnClickListener
            }

            val gender = if (rbMale.isChecked) "Male" else "Female"   // Radio

            val hobbies = mutableListOf<String>()                     // CheckBox
            if (cbReading.isChecked) hobbies.add("Reading")
            if (cbMusic.isChecked) hobbies.add("Music")
            val hobbyText = if (hobbies.isEmpty()) "None"
                            else hobbies.joinToString(", ")

            val country = spCountry.selectedItem.toString()           // Spinner
            val notify = if (switchNotify.isChecked) "ON" else "OFF"  // Switch

            progressBar.visibility = View.VISIBLE                     // ProgressBar dikhao
            tvResult.text = "Please wait..."
```

```
            // 1.5 second baad result dikhao aur ProgressBar chhupa do
            Handler(Looper.getMainLooper()).postDelayed({
                progressBar.visibility = View.GONE
                tvResult.text = "Result:\n" +
                    "Name: $name\n" +
                    "Gender: $gender\n" +
                    "Hobbies: $hobbyText\n" +
                    "Country: $country\n" +
                    "Notifications: $notify"
            }, 1500)
        }
    }
}
```

Output: User ne naam Ali likha, Male, Music, Pakistan, notifications ON rakhe aur SUBMIT dabaya. Pehle ProgressBar ghoomta hai aur 'Please wait...' aata hai, 1.5 second baad: **Result: Name: Ali, Gender: Male, Hobbies: Music, Country: Pakistan, Notifications: ON**.

Note: ImageView ko sirf dikhane ke liye Kotlin ki zaroorat nahi, XML ki src hi kaafi hai. Kotlin mein usay tab pakro jab image badalni ho: imgLogo.setImageResource(R.drawable.ic_launcher_foreground).

5. Code kaise kaam karta hai (qadam ba qadam)

Qadam	Kya hota hai
1	App khulti hai, onCreate chalta hai, **setContentView** XML ko screen par laga deta hai (sab 9 views nazar aate hain).
2	**findViewById** se har view ko Kotlin ke variable mein pakra, taake un se baat kar saken.
3	Spinner ko mulkon ki list di (**adapter** se).
4	User Submit dabata hai to **setOnClickListener** ke andar ka code chalta hai.
5	EditText se **.text.toString()**, Radio/CheckBox/Switch se **.isChecked**, Spinner se **.selectedItem** padha.
6	ProgressBar **VISIBLE** kiya (ghoomne laga), kaam ke baad **GONE** (chhup gaya).
7	Sab values **tvResult.text** mein likh kar dikha di.

6. Aisi screen ka code likhne ka tareeqa (paper ke liye)

Kisi bhi sawal mein jab bohat saare views ek screen par maange jayein, to har baar yehi **7 qadam** follow karo:

Qadam	Kya karna hai
1. Sketch	Copy par screen ka chhota sa naksha banao aur views ko upar se neeche number do.
2. Root layout	XML ke shuru mein LinearLayout (vertical) likho. Views zyada hon to bahar ScrollView lagao.
3. Views likho	Naksha ki tarteeb se har view ka tag likho. **Har view ko id** do, width/height do, text/hint do.
4. Kotlin: setContentView	onCreate mein pehle setContentView(R.layout.activity_main).
5. Kotlin: findViewById	Jin views se kaam lena hai un sab ko findViewById se pakro. (TextView/ImageView jo sirf dikhane ke liye hain unhein pakarna zaroori nahi.)

Qadam	Kya karna hai
6. Spinner ka adapter, button ka listener	Spinner ko adapter do. Button par setOnClickListener lagao.
7. Click ke andar	Values padho (.text.toString(), .isChecked, .selectedItem), logic lagao, result TextView par dikhao.

Id ke naam rakhne ka asaan usool (prefix)

View	Prefix	Example
TextView	tv	tvTitle, tvResult
ImageView	img	imgLogo
EditText	et	etName
RadioGroup / RadioButton	rg / rb	rgGender, rbMale
CheckBox	cb	cbReading
Spinner	sp	spCountry
Switch	switch (ya sw)	switchNotify
Button	btn	btnSubmit
ProgressBar	progressBar (ya pb)	progressBar

Aam ghaltiyan jinse nambar kat jate hain

(1) XML ki id aur Kotlin ke R.id.xxx ka naam alag likh dena. **(2)** RadioButtons ko RadioGroup mein na rakhna, to dono ek saath chun lete hain. **(3)** Spinner ko adapter na dena, to list khali. **(4)** EditText se seedha .text lena, .toString() likhna bhool jana. **(5)** Import lines likhna bhool jana. **(6)** RadioGroup mein dono RadioButtons par checked=true likhna (sirf ek par likho). **(7)** ProgressBar ko hide na karna ya shuru mein gone na rakhna.

Yaad rakho: Pehle XML ki id sochkar likho, phir Kotlin mein bilkul wohi naam R.id.idName likho. Yeh ek jaisa mel khana is subject ka sab se bara usool hai.